TRACK 2 / FOOD SECURITY DELIVERY OPERATIONS

Mealflo Hackathon Writeup

A working coordination app for turning intake, inventory, volunteer availability, and delivery constraints into safer food support routes.

PROBLEM

Track chosen and problem tackled

We chose Track 2: Food Security Delivery Operations. We tackled the coordination problem faced by volunteer-led food delivery programs, where incoming food requests, volunteer availability, meal inventory, routes, client needs, allergies, cold-chain requirements, accessibility constraints, and delivery status all have to be managed at the same time.

Our focus was helping a frontline coordinator turn scattered intake, inventory, and volunteer information into safe, assigned delivery routes with less manual work and fewer missed details - supported by a guided delivery system for volunteer drivers.

WORKING WEB APP

What we built

We built Mealflo, a web application for food support delivery operations. It connects public intake, admin coordination, and a volunteer driver experience, backed by a database layer, route calculation, map-based coordination, and language-model-assisted processing for intake and inventory workflows.

Public intake

Clients request food support and volunteers sign up to deliver meals. The system captures address, meal count, timing, dietary needs, allergies, availability, vehicle access, cooler capacity, and mobility constraints.

Admin operations dashboard

Coordinators review incoming requests, manage client records, view inventory, create routes, assign drivers, and track progress from one operational picture of what needs to happen today.

Volunteer driver view

Drivers use a phone-friendly interface to see assigned routes, follow directions one stop at a time, view delivery notes, notify clients, and mark deliveries complete.

OPERATOR WORKFLOW

How a frontline operator would use it on a real Monday morning

The workflow is intentionally low-cognitive-load: review requests, check constraints, assign routes, load meals, guide drivers, and track completion.

1 Review recent requests

Instead of searching through emails, forms, spreadsheets, and messages, the coordinator sees intake in one place with address, quantity, urgency, dietary needs, allergies, mobility needs, and delivery notes.

2 Check available inventory

They see which meals or staples are ready for delivery. Entries can be made manually or through uploading a receipt, with meals tagged by allergens or dietary restrictions.

3 Review volunteer availability

Mealflo captures driver constraints such as available time, starting location, vehicle access, cooler capacity, stair comfort, and other limitations.

4 Review generated routes

Deliveries are grouped into routes that consider urgency, driver capacity, and food requirements. The coordinator assigns a driver and confirms meals to load.

5 Guide the driver

The driver mobile view shows the next stop, address, meals to deliver, delivery notes, and a simple completion button.

6 Track completion

As stops are completed, the dashboard updates so the operator can see what is out for delivery, complete, and needing follow-up without calling every driver.

LIVE MAP

Delivery coordination snapshot

0 active

Updates as stops complete

READY TODAY

Coordinator view

Client	Urgency	Meals
Mabel Hart	9	2
Keisha Noor	9	2
Noel Okafor	9	3
Priya Sandhu	9	3

View all

Intended result

A lower-cognitive-load operating rhythm for frontline food support: every request, constraint, route, driver update, and follow-up action is visible in one place.

NEXT RELEASE

What we would ship next with one more week

With one more week, we would focus on expanding communication channels and improving real-world delivery coordination.

01

Connect additional intake sources

Many food support requests do not arrive through a formal web form. They may come through Instagram, Facebook, text message, or direct messages to the organization. We would add these as intake channels so Mealflo can capture requests from the places people already use.

02

Add real text messaging

Drivers need simple communication tools during delivery. We would add real text messaging from the driver flow, so a driver could send a genuine SMS through the app using a pre-written message such as: "Hi, this is your meal delivery driver. I'm outside now."

03

Get field data from a real organization

Finally, we would get field data from a real organization. That would let us test Mealflo against real intake patterns, delivery constraints, volunteer workflows, and communication needs, so the next version is shaped by frontline operations rather than guided and educated assumptions.

MEALFLO IN ONE LINE

A practical coordination layer for food support delivery teams: intake becomes structured, constraints become visible, routes become assignable, and drivers get guided execution on the phone.

Requests – Routes – Delivery